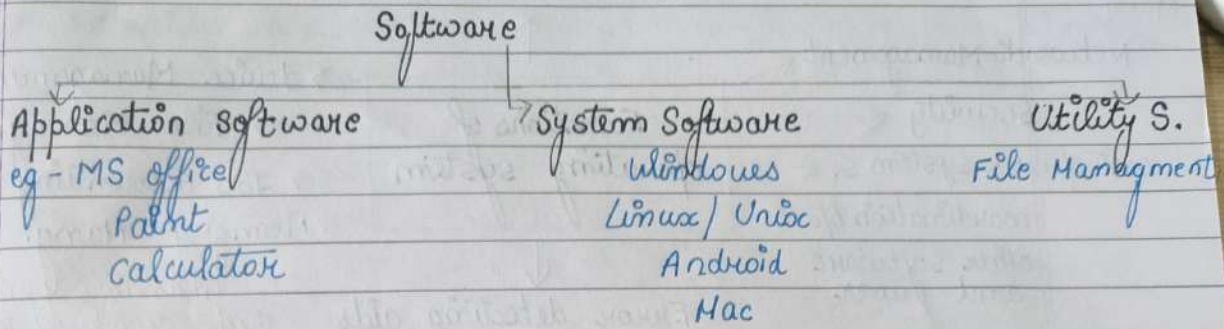


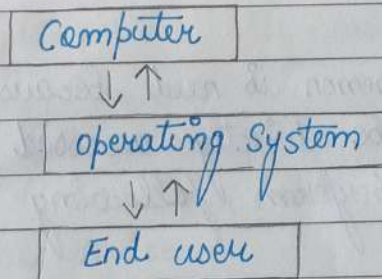
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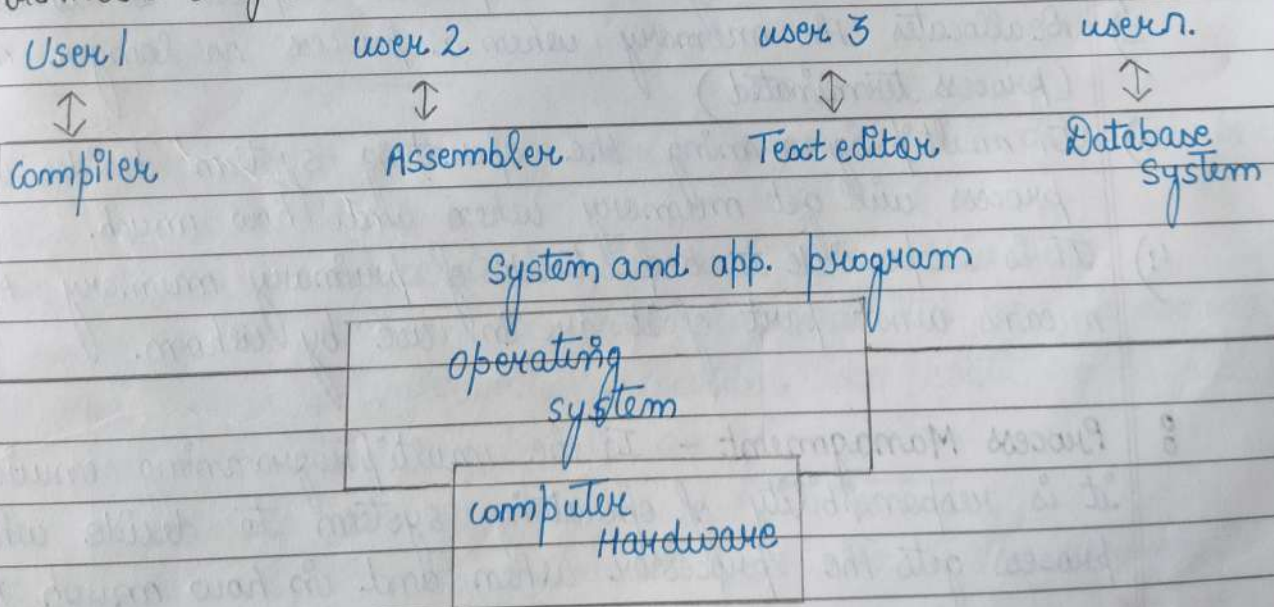


→ is a system software operating system that acts as a intermediary between user of computer and user and computer hardware that manages.

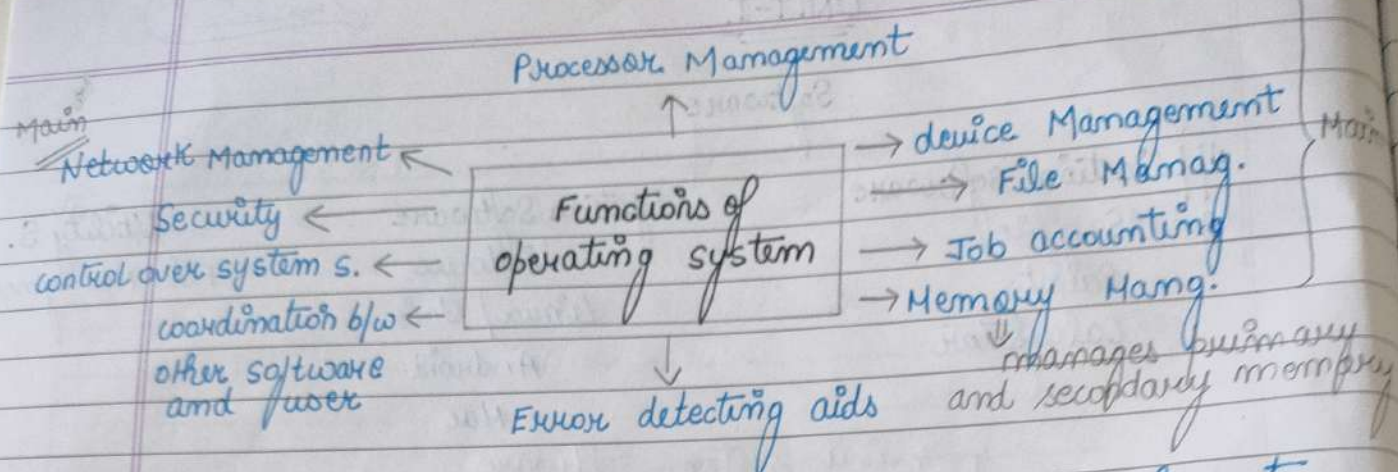
◉ Block diagram of OS -



◉ Architecture of OS -



◉ Functions of operating system -



◉ **Memory Management** - Memory Management refers to management of primary memory or ^{as we know that} Main Memory is a large array of words or bytes where each byte has its own address so we will need to manage it.

The above phenomenon is need because main memory is a fast storage that can be directly accessed by the CPU. operating system perform perform following activities for Memory Management -

- 1) Allocates the memory when a process requests it to do.
- 2) Deallocates the memory when a process no longer needs. (process terminated)
- 3) In multi programming the operating system decides which process will get memory when and how much.
- 4) It is responsible to keep track of primary memory that means what part of it are in use by whom.

◉ **Process Management** - In the multi programming environment it is responsibility of operating system to decide which process gets the processor when and in how much time.

The above function is called process scheduling.

The operating is also responsible for performing the following activities of Process Management -

- 1) Allocate the processor (CPU) to process whenever it needs.
- 2) Deallocates the processor to the process when it gets terminated.
- 3) It also keeps track of processor and status of process.

The program that is responsible for above task is known as traffic controller.

◦ Device Management - operating system manages the device communication through their respective drivers. operating system perform the following activities for device Management -

- 1) It is responsible for deciding which process gets the device and how much time.
- 2) Allocates device in efficient way.
- 3) Deallocates device.
- 4) Also keep track of all devices.

Program that is responsible for above task is known as Input output controller.

◦ File Management - file is generally organised into directories for easy navigation and uses.

These directories may contain file and other directories.

operating system perform the following activities for file manag. -

- 1) It keeps track of information, location, uses status of the files. The collective facilities are known as file system.
- 2) It decides who gets the resources.
- 3) The operating system is responsible for allocate the resources to the file.
- 4) Deallocation of the resources to the files is also responsibility of the operating system.

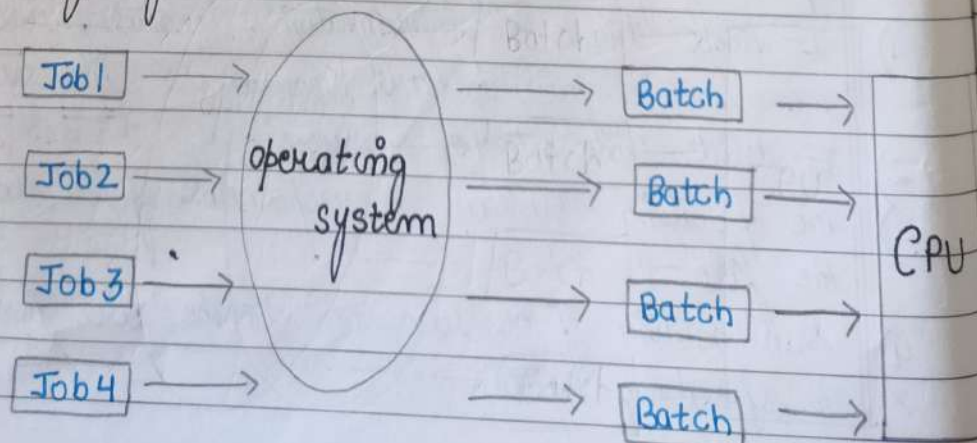
Apart from the basic function of OS some important activities are also performed by OS.

- (1) Security - OS prevents unauthorized access to program and data by providing the password and other similar security technique.
- (2) Control over system - OS is responsible for recording delay between request for a service and response from the system.
- (3) Job accounting - It is responsibility of OS to keep track of time and resources used by various jobs or user.
- (4) Error Detection - It is the responsibility to produce error message and other debugging and error detecting aids.
- (5) Coordination b/w other software and user - OS is also responsible for coordinating with compiler, interpreter, assembler and other software to the various users of the computer system.

§ Classification / Types of OS -

- | | |
|----------------------------|-----------------------|
| (1) Batch operating system | (5) Multiprocessor OS |
| (2) Interactive OS. | (6) Multiuser OS |
| (3) Time sharing OS | (7) Multiprocess OS |
| (4) Real Time OS | (8) Multithreaded OS |

Batch operating System -



1) Batch OS - In the Batch OS jobs were executed in Batches by using Batch OS users do not directly interact with a computer. Each and every user prepare their job using an offline devices like punch card etc and submitting it to computer operator.

Jobs with similar requirement are grouped and executed as a speed up processing. A computer system with batch OS perform the following activities -

- (i) Processing takes place in the order in which they are executed [FCFS - first come first serve]
- (ii) A job is a single unit that consists of a presence sequence of command data and program.
- (iii) These jobs are stored in a memory and executed w/o the need for manual information.
- (iv) When a job is successfully runs the OS release its memory.

2) Interactive OS - Interactive OS are computers that accept human input. Users gives commands or some data to the computers by typing or gestures. MS word, MS excel are eg of interactive system. Mac, Windows OS are eg of Interactive OS.

Definition - An interactive OS is an OS that allows execution of interactive programs. All PC [Personal Computers] are interactive OS. In an interactive OS the user enters commands into the system and work of the system is to be executed. In Anam - interactive programs Once started continuous w/o the need of human interaction. eg - compiler

3) Time Sharing OS - It allows multiple users or processes to

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continuously share the same system resources. This OS uses CPU scheduling and multi-programming to provide each user with a small portion of shared computer at once.

In this OS each and every user having an atleast one separate program in memory. When a program is loaded into memory then it is ready to execute, it performs a short period of time either completion or to complete input output.

⇒ Time slice / Time slot / Quantum - The short period of time during which the user gets the attention of CPU.
{value - 10 to 100 milliseconds.}

4) Real Time OS [RTOS] - It is an OS ^{in which} strict time constraints for any job to be performed. Whenever an event has occurred then it communicates to the comp. with help of sensors to monitor the event. It employs special purpose OS because conventional OS do not provide such performance.

eg - HTS, Linux, QNX, VX Works

Applications - It is used in real time app that work in specific guidelines.

- (i) It is used inside the Radar gadgets.
- (ii) It is used in inventory trading.
- (iii) Used inside cell phone switching, medical imaging system, fuel injection gadgets.

⇒ [Types - Hard, Soft, Firm]

(a) Hard Real Time OS - In this RTOS all critical task must be completed within the specified time duration of the system.

not meets the deadline, would result in critical failure such as damage to equipments or even loss of human life. eg - airbag provided by car-makers.

- (b) Soft Real Time OS - These OS accept some delay, which are accepted via the means of OS.

For eg - There may be a closing day assigned for a particular job but a delay for a small amount is acceptable so cutoff dates are treated softly.

These type of OS is used in transaction system and?

Ques - write diff b/w hard and soft real time OS?

- (c) Firm Real Time OS - In these type of OS we additionally want to absorb the deadline.

Advantages of RTOS -

- (1) By using this layout develop and execution of Real Time applications are so easy.
- (2) The working structure of these OS are extra compact. So these requires much less memory space.
- (3) In this type of OS max. usage / utilization of devices and systems.
- (4) The size of the program is small hence RTOS can also be embedded system, like in transperts and others.
- (5) In these type of OS men
- (6) These type of OS are error free.

Disadvantages of RTOS -

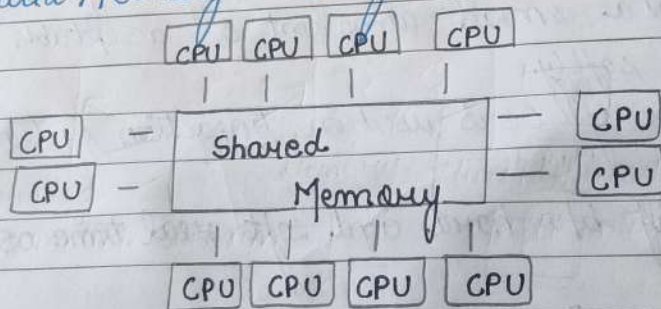
- (1) It is very costly to develop.
- (2) These type of OS consume critical CPU cycles.

5) Multiprocessor OS - when 2 or more CPUs share full access to common RAM, then these type of OS are called Multiprocessor OS. The main objective of this is to boost the system execution speed.

There are 2 types of Multiprocessor OS :-

(i) Shared Memory - All CPU share the common memory.

(ii) Distributed Memory - Every CPU has its own private memory.



Advantages of Multiprocessor -

- (1) In this system the workload can be distributed across multiple processors hence it can execute task faster than single processor.
- (2) An additional can be easily added to the system to handle increased work load.
- (3) If one processor fails then the remaining processors can continue executing the task.
- (4) These are cost effective.
- (5) Diff. processors can execute diff. tasks at a single time that's why parallelism will be enhanced.

Disadvantages of Multiprocessor -

- (1) These OS are more complex than single processor OS.
- (2) Requires more power than single processor OS.
- (3) Developing the software that can utilise multiprocessor

in single unit of time is challenging.

(6) **Multiuser OS** :- In a multiuser OS multiple users can access diff resources of a comp. system at the same time.

It have following types -

- (i) Distributed
- (ii) Time Sliced / Time sharing
- (iii) Multiprocessor

Advantages of Multiuser -

- (1) It helps in sharing of data / info / hardware resources etc.
- (2) If any one comp. fails it does not affect any other computer.
- (3) Backing up of data can be done easily.
- (4) In these type of OS users can share their work with other users

Disadvantages of Multiuser -

- (1) It requires expensive hardware to setup.
- (2) When multiple users log on or work on same system then it reduces overall performance of system.

(7) **Multi threading OS** -

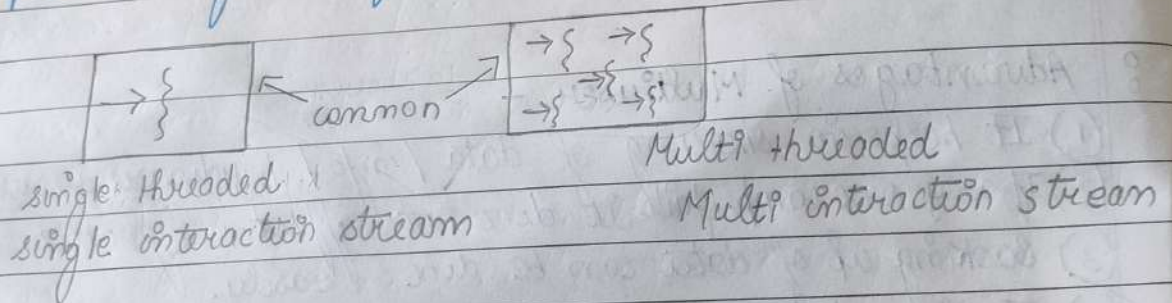
(#) **Thread** - Thread is a path which is followed during program execution.

(#) **Multitasking** - When 2 or more task executed at the same time or simultaneously these type of concept is Multitasking. It is categorised in 2 categories -

(i) Processor based

(ii) Thread based

- (#) **Process** - A process is a program being executed. Process can be further divided into independent units known as Threads.
- (#) **Thread** - A Thread is a small light weight process within a process or we can say a collection of thread is known as Process.
- A process may be single or multithreaded process.



8 Applications -

- (1) The concept of Threading is widely used in internet where we are using transaction processing of every type like Recharge, online transfer and banking etc.

By using the concept of threading less burden on CPU memory because the code is divided into small parts that are very light.

By using the concept of threading programming capability will be enhanced.

Life Cycle of a Thread -

- ① New
- ② Runnable
- ③ Time Waiting
- ④ Waited Time
- ⑤ Terminated

Thread goes mainly in 5 stages.

- (1) **New** - The life cycle of a born thread: (new thread) starts in this state. It remains in this state till the program starts.
- (2) **Runnable** - Thread become runnable after it starts. It is considered to be executing the task given to it.
- (3) **Waited time** - While waiting for another thread to perform a task the currently running thread goes into waiting state and transaction back again receiving the signal from other end.
- (4) **Time waiting** - A runnable thread enters into this state for a specific time interval and then transaction back when time interval expires.
- (5) **Terminated** - A thread enters into this state after completing its task or execution.

Types of Execution in OS -

Parallel Execution

Concurrent Execution

- (1) **Parallel** - This type of execution occurs whenever every thread in the process runs on a separate processor at the same time and in same multithreaded process.
- (2) **Concurrent** - This occurs when a processor is successful in switching resources b/w threads in a multithreaded process on a single processor.

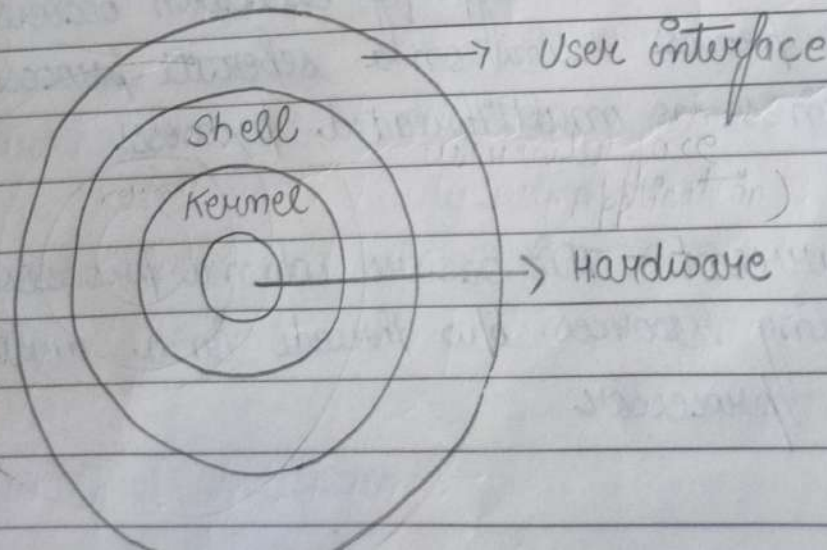
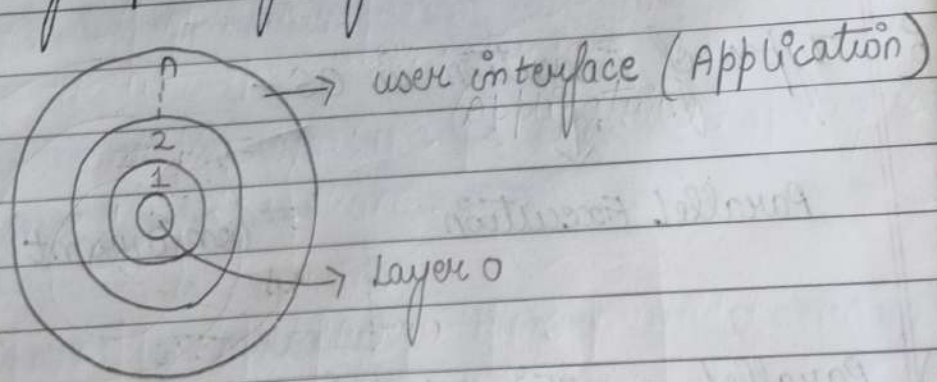
Drawbacks of Multithreading -

- (1) Data inconsistency and dead lock.
- (2) Thread starvation - Many thread try to access the same data.
- (3) Display issue occur if threads lack coordination.

Benefits of Multithreading -

- (1) It can improve the performance and efficiency of a program by utilising available resources.
- (2) It can enhance responsiveness in application that also involves user interaction.
- (3) It can enable better resource utilization, better code organization and modularity by dividing complex task into smaller.

⇒ Layered structure of operating system -



In layered approach OS is broken into no. of layers (levels). The bottom layer (layer 0) is a hardware layer and the highest layer (layer N) is responsible for user interface.

OS layer is an implementation of an abstract object made up of data and the operations that can manipulate these data. In

OS layer architecture any layer M consists of data structure and set of routines that can be invoked by higher layer.

Each layer is implemented with only those operations provided by lower level layers that means a particular layer can access all the layers present below it but it cannot access them above layer, for eg- layer N-1 can access all the layers from N-2 to 0 but cannot access the Nth layer.

Layer 0 deals with allocating the process, switching b/w process when interaction occurs or time expires it also deals with the basic Multiprogramming of CPU.

Advantages of layered operating system -

Layer No	Function
5	the operator
4	user program
3	I/O
2	operator process communication
1	Memory Management
0	Processor & Management

Modularity - Each layer perform only the task it is scheduled to perform.

Easy debugging - Layer structure is discrete, hence it is very easy to debug any type of error.
eg - if an error occurs in the CPU scheduling layer so

- 3) Easy update - Modification made in a particular layer will not affect other layers.
- 4) No direct access to Hardware - Hardware layer is the inner most layer of OS layered structure for that reason user can use the services of Hardware but cannot directly modify or access it.
- 5) Abstraction - Every layer having its own function and the functions and implementations of the other layers are abstract to it.

Disadvantage -

- 1) Complex and careful implementation - A layer can access the services of the below layer so the arrangement of layers must be done carefully.
- 2) Slower in execution - If a layer wants to interact with other layer then it sends a request that has to travel to all the layers present in b/w the 2 interacting layer. Hence it increases response time.

Shell

Shell - Shell is an environment which provides an interface to user to use or OS services.

Kernel - Kernel is the heart or core of an OS that manages operations of computer and hardware. Basically acts as a bridge b/w the user and the resources of the system by accessing different computer resources (CPU, I/O device).

Difference b/w shell and kernel -

Shell

- 1) It allows the user to communicate with the kernel.
- 2) It is an interface b/w kernel and user.
- 3) Shell is the command line interpreter (CLI).
- 4) Its types are -
 - (i) Bourne Shell
 - (ii) C Shell
 - (iii) Korn Shell
- 5) It carries out commands on a group of files by specifying a pattern to match shell command like ls, mkdir and many more commands can be used to request to complete the specific operation to the OS.
- 6) It is the outer layer of OS.
- 7) It interacts with user and interprets to machine understandable language.
- 8) It executes commands and programs.
- 9) It provides various features like tab completion, scripting capability and command history.
- 10) Commands are executed on a

Kernel

- 1) Kernel controls all the tasks of the system.
- 2) It is the core of the OS.
- 3) It is a low level program interfacing with hardware on the top of which applications are running.
- 4) Its types are -
 - (i) Monolithic Kernel
 - (ii) Micro Kernel
 - (iii) Hybrid Kernel
- 5) It is responsible to perform Memory Management and Process Management.
- 6) It is the inner layer of OS.
- 7) Kernel directly interacts with hardware by accepting machine understandable language from shell.
- 8) It enables user and applications to interact with H/W resources.
- 9) It is responsible for tasks like Memory Management, process scheduling and device drivers.
- 10) The memory is managed by

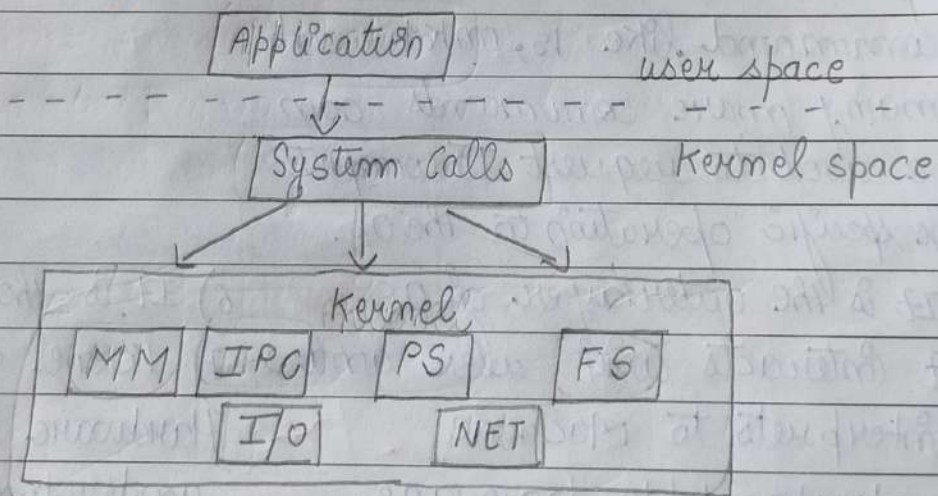
File collection by giving a pattern to match.

⇒ Architecture of operating system - It can be of 4 types

- (i) Monolithic
- (ii) Layered
- (iii) Micro kernel
- (iv) Hybrid

(i) Monolithic - In Monolithic architecture each component of OS is contained in the kernel that means it works in kernel space and the component of the OS communicate with each other using function calls.

Eg of this type of architecture is VMX, Linux



Advantages

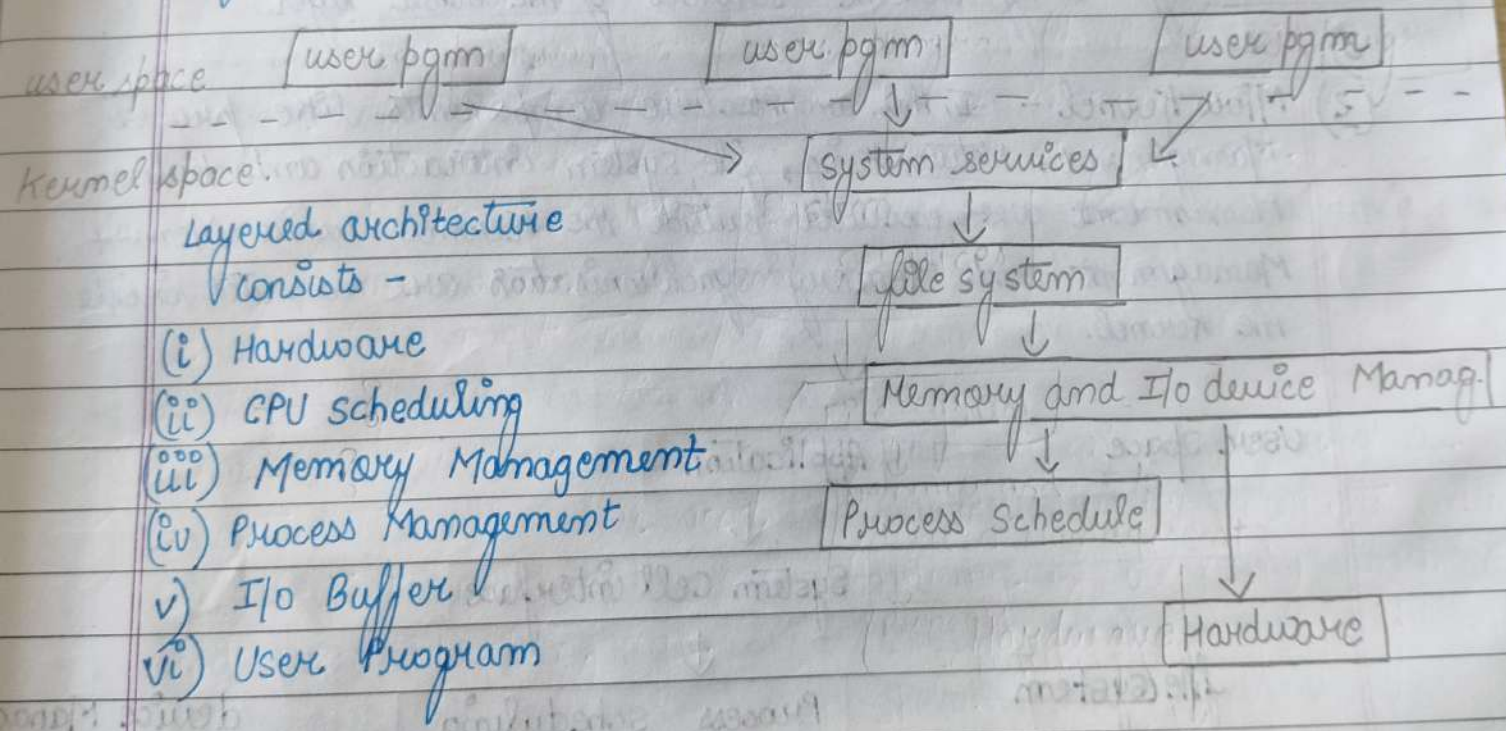
- (1) It provides CPU scheduling Memory Management via system call.
- (2) In a single process space the entire process is running.
- (3) It is a single static binary file.

Disadvantages -

- (1) In the architecture all the components are interdependent, if one component fails then entire system will fail.
- (2) If a user adds new services then the entire OS needs to be changed.

2) Layered - In layered architecture components with similar functionalities are grouped to form a layer, according to this if $n+1$ layers are constructed then it will be counted from 0 to n . Each and every layer has different functionality and services.

Eg - Windows XP, Linux



Hardware

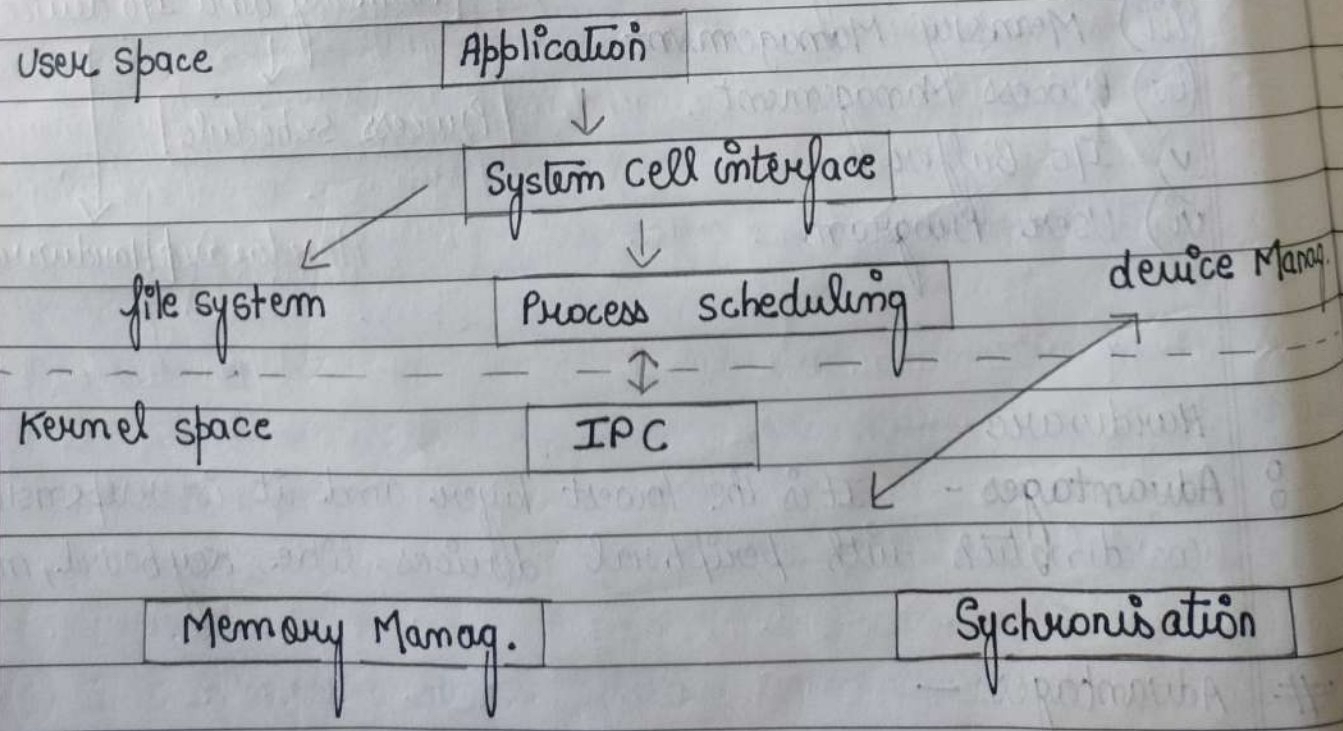
- It is the lowest layer and it is responsible for coordination with peripheral devices like keyboard, mic, speaker.

Advantages -

- (1) It provides Modularity.
- (2) It has independent component so changing one of them will not affect the other.
- (3) User can access only the services of hardware layer, it cannot access the Hardware layer access.
- (4) Each layer has its own functionality.

Disadvantages -

- 1) If one layer wants to communicate with other layer then it goes through all layers b/w that, that will ↑ response time efficiently.
- 2) The implementation of layered architecture is complex because one layer may use the services of the other layer.
- (3) **Micro Kernel** - In this architecture components like process Management, Networking, file system interaction and device Management are executed outside the Kernel while Memory Management and Memory synchronisation are executed inside the Kernel.



- 1) The processes that are inside the kernel having higher priority as compared to the components that are outside the kernel.
- 2) In this architecture components possess higher modularity, Hence if one or more components fails then system keeps working.

Advantage -

- 1) These type of OS are modular so when we distribute one of the component then it will not affect other components.
- 2) Adding of new feature can be done easily.
- 3) This architecture is compact.

Disadvantage -

- 1) If we need to implement a driver as processors then it will need function call.
- 2) Services provided by this architecture is costlier than Monolithic OS.

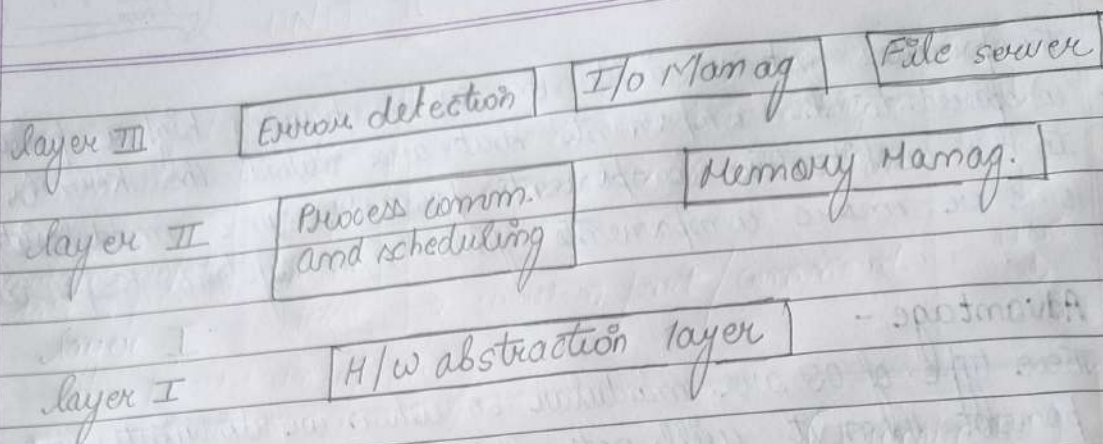
4) Hybrid - It consists of combining any [Micro Kernel, Monolithic or Layered] architecture all together.

-: Hybrid architecture consists of 3 layers -

(i) Hardware Abstraction Layer - This is the lowest level. This layer provides an interface b/w Hardware and Kernel.

(ii) Micro Kernel Layer - This layer consists of CPU scheduling, IPC [Interprocess Communication], Memory Management.

(iii) Application Layer - This layer provides the functionality like error detecting, I/O device Management, file server. This layer acts as an interface b/w Micro Kernel and user.



Advantage -

- 1) It is easy to manage because this architecture uses a layered approach.
- 2) Security and protection are relatively improved.
- 3) Number of layers is lesser.
- 4) This architecture allows various architecture to provide their services respectively.

Disadvantage -

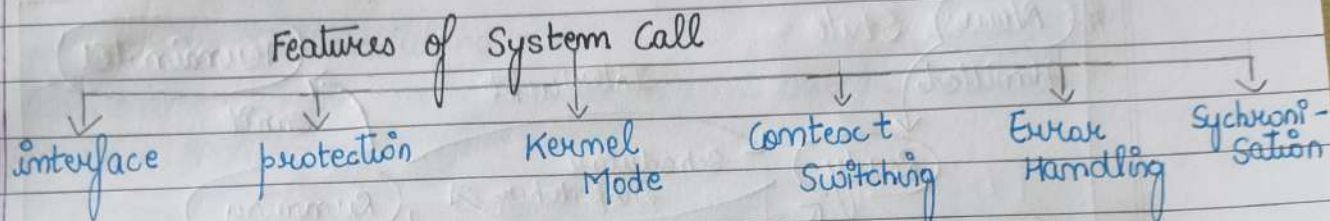
- 1) Hybrid architecture keeps certain services in kernel space while moving less critical services to user space.

o System Calls - It is a programmatic way in which a computer program requests a service from the kernel of the OS, it is executed on. Hence we can say that a system call is a program to interact with the OS. Computer program makes a system call when it makes a request to the OS. Kernel system call provides an interface b/w a process and an OS to allow user level processes to request services of OS.

⇒ Services provided by System Call -

- ① Process Creation
- ② Process Management

- ③ Main Memory Management
- ④ File Access
- ⑤ Directory and File System Management
- ⑥ Device Handling (I/O)
- ⑦ Protection
- ⑧ Networking



⇒ **Advantage of System Call -**

- ① Access to Hardware resource - by using system call to allow access
- ② Memory Management - (allocation and deallocation) to all H/w resources.
- ③ Process Management - it is the responsibility of system call to create or access a process.
- ④ Security
- ⑤ Standardisation [8.4 / 8.5] - *Process / Thread / Shared Library / Shared Object*

⇒ **How does System Call Work -**

- ① user needs special resources
- ② Program makes a system call request (to kernel)
- ③ OS sees the system call
- ④ OS perform the operation
- ⑤ OS give control back to program.

eg of System Call -

Process	① Process Control	create process() Exit process() wait for single object()
	② File Manipulation	create file() Read file() Write file()